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KOLKATA

BACHELOR OF STATISTICS (B.STAT. HONS.)

Statistical Methods III — Semester Project



Customer Segmentation using Gaussian Mixture Models via the EM Algorithm

A Model-Based Clustering Approach on Online Retail Transactional Data

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Abstract

Customer segmentation is a foundational task in commercial data analytics, helping businesses group buyers based on purchasing patterns. While standard algorithms like k-means are easy to run, they assume that clusters are spherical, well separated and equal in size—an assumption that rarely holds for real-world purchase data. In this study, we apply Gaussian Mixture Models (GMM) estimated via the Expectation-Maximization (EM) algorithm to segment customers in an online retail as an improvement of the widely used traditional cell based RFM clustering. We aggregate raw purchase logs into Recency, Frequency, and Monetary (RFM) variables and apply a log transformation to handle heavy right-skewness. Using the Bayesian Information Criterion (BIC), we determine the optimal number of clusters and covariance structures. The resulting probabilistic clusters give us clear, interpretable customer groupings that offer useful practical guidance for target marketing.

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1 Introduction

1.1 Motivation and Framework

Modern e-commerce platforms generate massive line-item transaction logs detailing every product purchased. To translate these raw logs into actionable commercial insights, retail businesses group buyers into behavioral segments using **RFM Analysis**:

- **Recency (R):** Days elapsed since the customer's last order.
- **Frequency (F):** Total number of distinct orders placed.
- **Monetary Value (M):** Total expenditure across all purchases.

While k-means is commonly used for RFM clustering, it assumes rigid spherical clusters of equal variance. Retail spend data exhibits heavy right-skewness, non-spherical covariance shapes, and overlapping segment boundaries. We address these limitations using probabilistic **Gaussian Mixture Models (GMMs)** optimized via the **Expectation-Maximization (EM) algorithm**.

1.2 Dataset Overview and Processing Pipeline

We analyze an online retail transactional dataset containing item-level entries with invoice codes, stock descriptions, quantities, unit prices (in GBP), timestamps, and customer identifiers. Before clustering, we filter out cancelled orders, missing customer IDs, and non-retail bulk purchases.

Our methodological workflow is executed across six key stages:

1. **Data Cleaning:** Filter out cancelled and incomplete records[T1],[T2].
2. **Exploratory Analysis:** Inspect geographic distributions, sales seasonality, and wholesale outliers[T3],[T4],[T5].
3. **Feature Engineering:** Calculate customer RFM vectors, apply natural log scaling, and standardize features[T6],[T7],[T8].
4. **Model Estimation:** Fit candidate GMMs across components $G \in \{1, \ldots, 20\}$ and select the best model using BIC[T9],[T10].
5. **Cluster Profiling:** Assign MAP segment labels and compute unscaled summary metrics[T11],[T12],[T13].
6. **Commercial Strategy:** Translate profile characteristics into targeted business recommendations.

2 Data Pipeline and Exploratory Analysis

2.1 Data Cleaning and Filtering

Raw retail logs require cleaning before customer-level features can be built[T1],[T2]. Our preprocessing routine consists of the following steps:

1. Remove records with missing `CustomerID` entries, as unauthenticated guest purchases cannot be tracked over time.

2. Filter out cancelled orders (invoices starting with 'C') and non-positive item quantities or prices (Quantity ≤ 0, UnitPrice ≤ 0).

3. Restrict transactions to regular retail orders.

2.2 Exploratory Data Analysis (EDA)

2.2.1 Geographic Distribution

We first examine how orders are distributed across different countries[T3]. As shown in Figure 1, the UK accounts for the vast majority of orders, with small contributions from European countries like Germany, France, and EIRE. Focusing primarily on UK buyers helps remove confounding variables caused by international shipping costs and different delivery timelines.

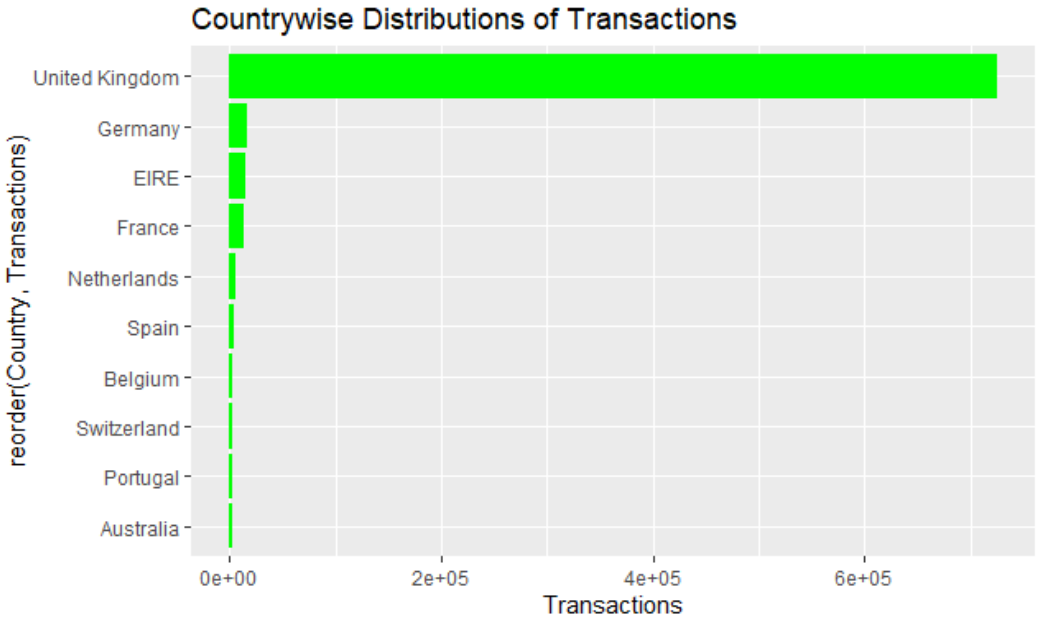


Figure 1: Geographic distribution of raw transactions across operating countries.

2.2.2 Monthly Revenue Dynamics

Analyzing monthly sales figures[T4] reveals clear seasonal patterning (Figure 2). Sales remain steady for most of the year before climbing sharply in November and December, reflecting standard Fourth-Quarter holiday purchasing.

2.2.3 Handling Wholesale Outliers

Extreme bulk transactions can distort cluster covariance estimates. To identify non-retail buyers, we isolate wholesaler volume thresholds[T5]. As visualized in Figure 3 and Figure 4, a few orders involve thousands of units per transaction. We apply upper threshold limits to isolate representative retail buyers.

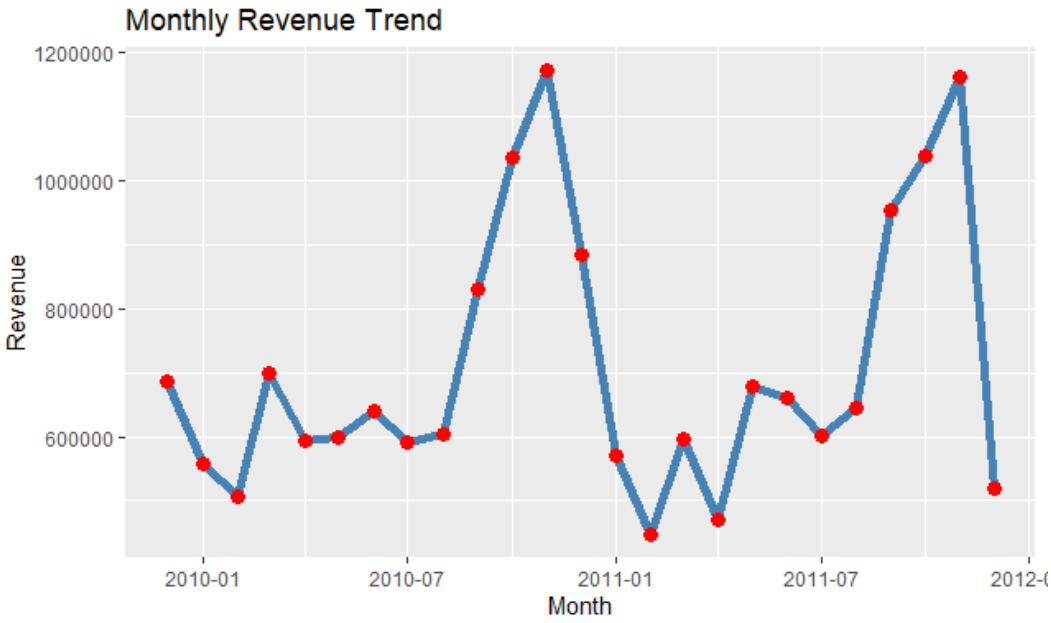


Figure 2: Monthly revenue trend highlighting e-commerce sales seasonality.

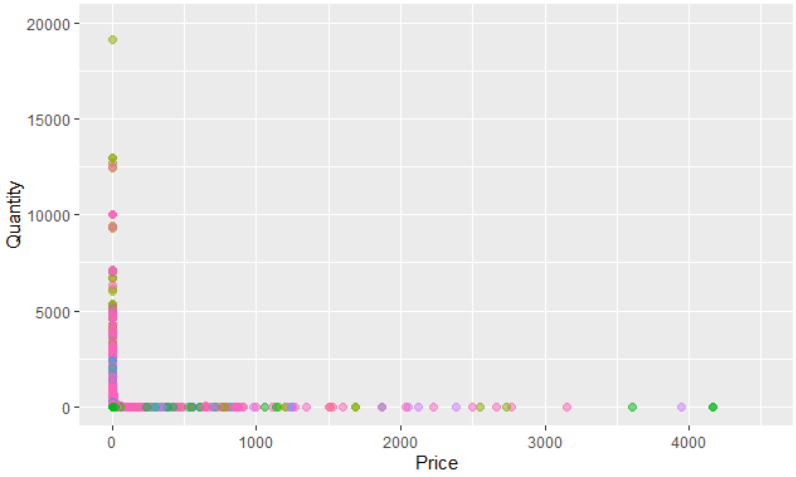


Figure 3: Wholesaler Threshold Isolation identifying extreme bulk non-retail buyers.

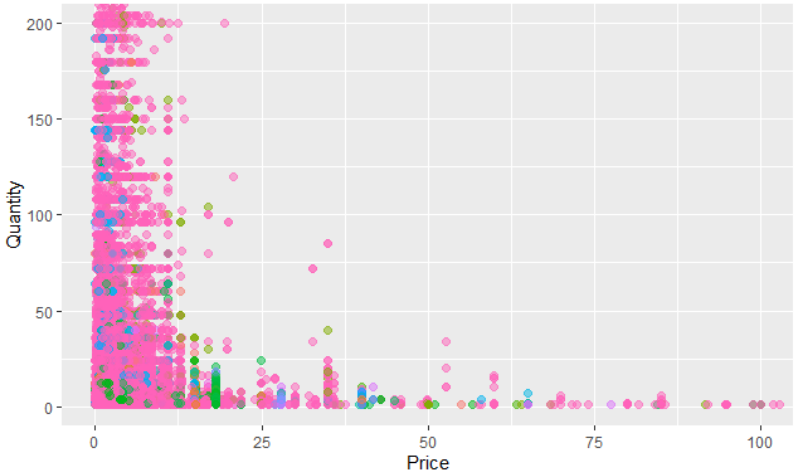
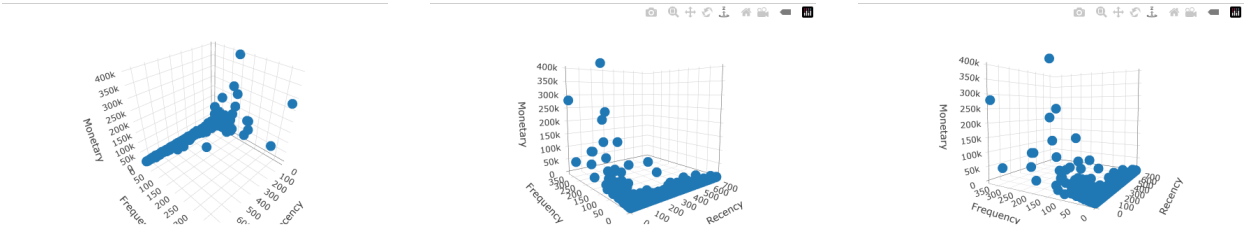


Figure 4: Progressive trimming plot displaying high-quantity outlier removal (Cutoff 100–200).

2.3 RFM Transformations and Normalization

After cleaning the item logs, we aggregate transactions by `CustomerID` to compute raw Recency (R), Frequency (F), and Monetary spend (M)[T6]. Because raw financial metrics are heavily right-skewed and concentrated near zero (Figure 5), fitting Gaussian distributions directly on raw metrics would violate normality assumptions.



(a) Raw RFM Angle 1 (b) Raw RFM Angle 2 (c) Raw RFM Angle 3

Figure 5: Raw 3D RFM distribution showing severe right-skewness and origin compression.

To resolve this, we apply natural log scaling followed by z-score standardization[T7],[T8]:

$$R'_i = \log(R_i + 1), \quad F'_i = \log(F_i + 1), \quad M'_i = \log(M_i + 1) \tag{1}$$

$$\mathbf{x}_i = \left(\frac{R'_i - \bar{R}'}{s_{R'}}, \frac{F'_i - \bar{F}'}{s_{F'}}, \frac{M'_i - \bar{M}'}{s_{M'}} \right)^T \tag{2}$$

As shown in Figure 6, this transformation successfully converts the skewed data into well-behaved, elliptical component shapes suitable for GMM modeling. A point to note is that the log transformation is done after adding one to avoid taking logarithm of 0. But for our data all values being strictly greater than 0, we ignored this safety measurement during code implementation.[T7]

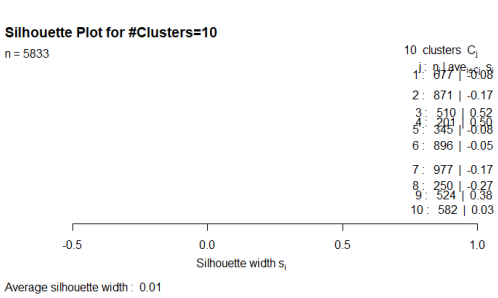


(a) Log-Transformed 3D View 1 (b) Log-Transformed 3D View 2

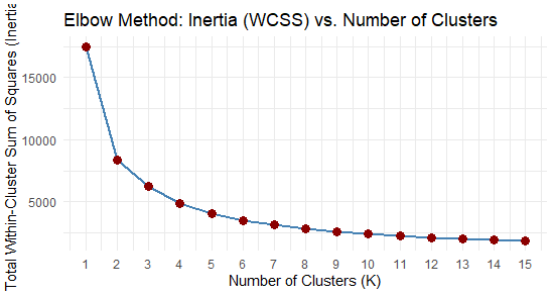
Figure 6: Log-transformed and standardized 3D RFM distribution displaying elliptical component shapes. Click here to view the Interactive Normalized Plot

3 Incompatibility of Hard Clustering

The Silhouette index being close to 0 implies that hard clusters are merely significant for this purpose. Additionally the sharp drop in slope of the Inertia plot at $K = 2$ indicates that there can



(a) Silhouette Index Output



(b) Elbow plot for Inertia

Figure 7: Hard Clustering Metric Plots

be only 2 significant clusters in this data. The reason for these poor performance of the metrics comes down to some fundamental assumptions during any standard clustering(such as k-means):

- The Clusters are Spherical.
- The Clusters are well separated from each other.
- The Clusters are roughly same in size.

As we shall see later, the second assumption is in fact responsible for such poor performance. The aforementioned discussion triggers us to adopt soft clustering using GMMs as a more sensible and prophylactic approach.

4 Theoretical Framework

4.1 Gaussian Mixture Models (GMM)

Let $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$ be a dataset of n independent, d-dimensional continuous observations. A Gaussian Mixture Model assumes that the marginal density of each observation $\mathbf{x}_i$ is a weighted sum of G multivariate normal distributions:

$$f(\mathbf{x}_i \mid \Theta) = \sum_{k=1}^{G} \pi_k \, \phi(\mathbf{x}_i \mid \mu_k, \Sigma_k) \tag{3}$$

where π_k represents the mixing proportion for the k-th component ($\pi_k > 0, \sum_{k=1}^{G} \pi_k = 1$), $\phi(\cdot \mid \mu_k, \Sigma_k)$ is the multivariate normal density with mean vector μ_k and covariance matrix Σ_k, and $\Theta = \{\pi_k, \mu_k, \Sigma_k\}_{k=1}^{G}$ denotes the complete set of parameters.

4.2 The Expectation-Maximization (EM) Algorithm

Directly maximizing the log-likelihood function for a mixture model is non-trivial because component memberships are unobserved. To handle this, we introduce latent indicator vectors $\mathbf{z}_i = (z_{i1}, \ldots, z_{iG})^T$, where $z_{ik} = 1$ if observation i belongs to component k, and 0 otherwise.

The EM algorithm updates parameter estimates iteratively:

- **E-Step (Expectation):** Calculate the posterior probability (responsibility) that observation i originates from component k:

$$\hat{\omega}_{ik}^{(t+1)} = \frac{\pi_k^{(t)} \phi(\mathbf{x}_i \mid \mu_k^{(t)}, \Sigma_k^{(t)})}{\sum_{j=1}^{G} \pi_j^{(t)} \phi(\mathbf{x}_i \mid \mu_j^{(t)}, \Sigma_j^{(t)})} \tag{4}$$

- **M-Step (Maximization):** Update the parameters using the responsibilities calculated in the E-step:

$$\pi_k^{(t+1)} = \frac{n_k}{n}, \quad \mu_k^{(t+1)} = \frac{1}{n_k} \sum_{i=1}^{n} \hat{\omega}_{ik}^{(t+1)} \mathbf{x}_i, \quad n_k = \sum_{i=1}^{n} \hat{\omega}_{ik}^{(t+1)} \tag{5}$$

$$\Sigma_k^{(t+1)} = \frac{1}{n_k} \sum_{i=1}^{n} \hat{\omega}_{ik}^{(t+1)} (\mathbf{x}_i - \mu_k^{(t+1)})(\mathbf{x}_i - \mu_k^{(t+1)})^T \tag{6}$$

- **Stopping Criterion:** Stop the iterations when

$$|\ell_{inc}(\Theta^{(t)}) - \ell_{inc}(\Theta^{(t-1)})| < \varepsilon$$

for some small ε(say 10^{-6}), where $\ell_{inc}(\Theta)$ denotes the incomplete log likelihood when the true parameter is Θ.

4.3 Covariance Parameterization and Model Selection

Following Banfield and Raftery (1993), the spectral decomposition of component covariance matrices Σ_k are written in the following manner:

$$\Sigma_k = \lambda_k \mathbf{D}_k \mathbf{A}_k \mathbf{D}_k^T \tag{7}$$

Here, λ_k controls the volume of the cluster, $\mathbf{D}_k$ determines its orientation, and $\mathbf{A}_k$ determines its shape. Conventionally we choose the scalar λ_k in such a way so as to ensure $\det(\mathbf{A}_k) = 1$. $\mathbf{A}_k$ is a diagonal matrix containing the eigenvalues of the variance-covariance matrix in descending order. By constraining or freeing these components, we evaluate spherical (`EII`, `VII`), diagonal (`EEI`, `VEI`, `EVI`, `VVI`), and ellipsoidal (`EEE`, `EEV`, `VEV`, `VVV`) models.

To choose the best model configuration without overfitting, we use the Bayesian Information Criterion (BIC):

$$\text{BIC} = 2 \cdot \ell(\hat{\Theta} \mid \mathbf{X}) - \nu \log(n) \tag{8}$$

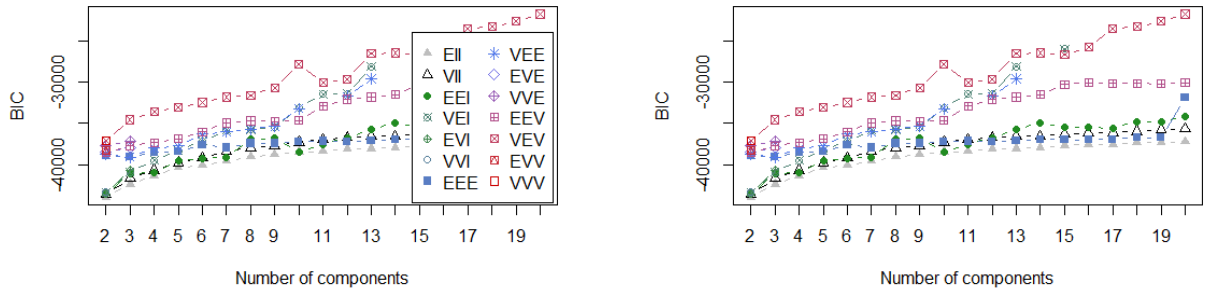
where $\ell(\hat{\Theta} \mid \mathbf{X})$ is the maximized log-likelihood, ν is the number of estimated parameters, and n is the sample size. Under the `mclust` convention in R, higher BIC values correspond to better-fitting models.

The BIC plot is suggestive of adopting `VEV` model (variable volume, equal shape, variable orientation) for our clustering problem.

4.4 Optimizing Covariance Matrix for VEV Models:

Though the updation rules for the responsibility, means remain the same, optimizing the covariance matrix under the shape constraint is demonstrated below.

$$\mathscr{Q}_\Sigma = -\frac{1}{2} \sum_{k=1}^{G} n_k \left[\log|\Sigma_k| + \frac{1}{n_k} \sum_{i=1}^{n} z_{ik}^{(t)} (x_i - \mu_k^{(t)})^\top \Sigma_k^{-1} (x_i - \mu_k^{(t)}) \right] \tag{9}$$



(a) BIC with legends for model selection

(b) BIC without legends

Figure 8: BIC plot for different number of clusters

We shall maximize (7) with respect to Σ_k. Note that $x^\top Mx = \mathrm{tr}(Mxx^\top)$ and writing (getting rid of $^{(t)}$) $W_k = \frac{1}{n_k}\sum_{i=1}^{n} z_{ik}^{(t)}(x_i - \mu_k^{(t)})(x_i - \mu_k^{(t)})^\top$ we get the following:

$$\min_{\Sigma}(-Q_\Sigma) = H(\Sigma_1,\ldots,\Sigma_G) = \sum_{k=1}^{G} n_k \left(\log|\Sigma_k| + \mathrm{tr}(\Sigma_k^{-1}W_k)\right)$$

As stated earlier, in VEV model, we write $\Sigma_k = \lambda_k D_k A D_k^\top$ and that $\det(A) = 1$ also D_k being orthogonal, has determinant 1. Essentially we have $\det(\Sigma_k) = \lambda_k^p$, for p-dimensional data(here p=3), or in other words, the determinant equals volume raised to number of dimensions. Now writing H as a function of the shape, volume and orientation, we get:

$$H(\lambda_k, D_k, A) = \sum_{k=1}^{G} n_k \left[p\log\lambda_k + \frac{1}{\lambda_k}\mathrm{tr}(D_k A^{-1} D_k^\top W_k)\right]$$

Optimization with respect to D_k**:** By the cyclic property of the trace,

$$\mathrm{tr}(D_k A^{-1} D_k^\top W_k) = \mathrm{tr}(A^{-1} D_k^\top W_k D_k)$$

. Let the spectral decomposition of the empirical scattering matrix be $W_k = U_k \Gamma_k U_k^\top$, where $\Gamma_k = \mathrm{diag}(\gamma_{k1},\ldots,\gamma_{kp})$ contains the eigenvalues in descending order, and U_k is the orthogonal matrix of eigenvectors. By the Von Neumann Trace Inequality, we find that the trace is minimized over different choices of the orientation matrix, when the orientation matrix aligns with the eigenvectors of the empirical data matrix,i.e. they have the same set of eigenvectors. $\boxed{D_k^{(t)} = U_k}$. (Note that A having its eigenvalues arranged in descending order implies that A^{-1} will have its eigenvalues in the ascending order, hence in opposite order to that of Γ_k). Substituting $D_k^{(t)}$ back, the trace becomes $\mathrm{tr}(A^{-1}\Gamma_k) = \sum_{j=1}^{p} \frac{\gamma_{kj}}{a_j}$.

Optimization with respect to λ_k**:** The objective function now depends only on volume and shape:

$$H(\lambda_k, A) = \sum_{k=1}^{G} n_k \left[p\log\lambda_k + \frac{1}{\lambda_k}\sum_{j=1}^{p}\frac{\gamma_{kj}}{a_j}\right]$$

Taking the partial derivative with respect to λ_k and setting it to zero:

$$\frac{\partial H}{\partial \lambda_k} = \frac{n_k p}{\lambda_k} - \frac{n_k}{\lambda_k^2}\sum_{j=1}^{p}\frac{\gamma_{kj}}{a_j} = 0$$

$$\boxed{\lambda_k^{(t)} = \frac{1}{p}\sum_{j=1}^{p}\frac{\gamma_{kj}}{a_j} = \frac{1}{p}\text{tr}(A^{-1}\Gamma_k)}$$

***Constrained Optimization with respect to A*:** Plugging back the value of $\lambda_k^{(t)}$ and simplifying, we get

$$H(A) = \sum_{k=1}^{G} n_k \log\left(\sum_{j=1}^{p}\frac{\gamma_{kq}}{a_q}\right)$$

getting rid of terms that do not depend upon A. We need to minimize $H(A)$ subject to the constraint $\prod_{j=1}^{p} a_j = 1 \iff \sum_{j=1}^{p}\log a_j = 0$, where a_js are the diagonal elements of the matrix A. We construct the Lagrangian with multiplier η:

$$\mathscr{L}(a_1,\ldots,a_p,\eta) = \sum_{k=1}^{G} n_k \log\left(\sum_{q=1}^{p}\frac{\gamma_{kq}}{a_q}\right) - \eta\sum_{j=1}^{p}\log a_j$$

.

$$\frac{\partial\mathscr{L}}{\partial a_j} = 0 \implies \frac{\partial\mathscr{L}}{\partial a_j} = \sum_{k=1}^{G} n_k \frac{-\gamma_{kj}/a_j^2}{\sum_{q=1}^{p}\gamma_{kq}/a_q} - \frac{\eta}{a_j} = 0$$

$$\iff \sum_{k=1}^{G} n_k \frac{\gamma_{kj}/a_j}{\sum_{q=1}^{p}\gamma_{kq}/a_q} + \eta = 0 \implies \sum_{j=1}^{p}\left(\sum_{k=1}^{G} n_k \frac{\gamma_{kj}/a_j}{\sum_{q=1}^{p}\gamma_{kq}/a_q}\right) + \sum_{j=1}^{p}\eta = 0$$

. Swapping the summation order and simplifying we get: $\eta = \frac{-n}{p}$. Substituting η back and writing the denominator in terms of $\lambda_k^{(t)}$, we get:

$$\sum_{k=1}^{G} n_k \frac{\gamma_{kj}/a_j}{p\lambda_k^{(t)}} - \frac{n}{p} = 0 \implies \boxed{a_j^{(t)} = \frac{1}{n}\sum_{k=1}^{G}\frac{n_k\gamma_{kj}}{\lambda_k^{(t)}}}$$

Motivation of Invoking EM Algorithm: That the updation rule of A depends on $\lambda_k^{(t)}$ which in turn depends on A suggests us that we might have to follow some iterative algorithm. Hence we use the EM algorithm.

5 Empirical Results and Model Selection

5.1 GMM Estimation and BIC Evaluation

We fit mixture models across components $G \in \{1,\ldots,20\}$ and standard covariance structures using the `mclust` package[T9],[T10]. With reference to the BIC plots in Model Selection we narrow down to 2 models of interest as following inTable 1.

The optimal GMM selection is identified by locating the local maximum point on the BIC curve (Figure 8), though the elbow occurs at $K = 13$. We sacrifice a bit of information for a bit of interpretation(We evaluated all the results for $K = 13$ as well). More details on the models can be found here.

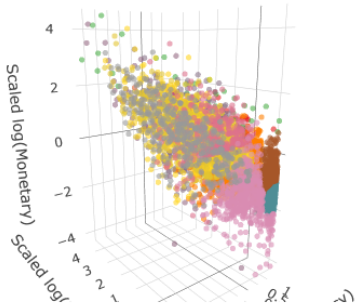
Table 1: Top Candidate GMM Models Ranked by BIC

Model Type	Covariance Structure	Clusters (G)	BIC Value
Elbow Model	VEV	13	-26592.28
Extrema Model	VEV	10	-26631.29

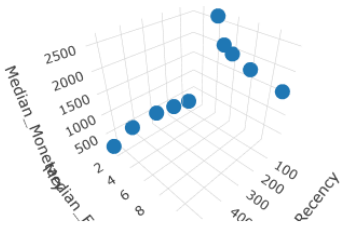
5.2 Post-Clustering Profiling

We assign each customer to a final segment using Maximum A Posteriori (MAP) classification ($\hat{z}_i = \arg\max_k \hat{\omega}_{ik}$)[T11],[T12],[T13]. Transforming the standardized metrics back to original scale yields the segment profiles summarized below and visualized in Figure 9:

Cluster	Customer_count	Median_Recency	Median_Frequency	Median_Monetary
5	345	33.90208333333333	4	2598.28
1	677	40.875	5	2047.51
7	977	40.99930555555556	6	1921.76
2	871	36.84027777777778	8	1727.43
8	250	58.49618055555554	11.5	1628.71
4	201	386.11041666666665	4	1059.41
6	896	214.87048611111112	2	530.565
9	524	423.4065972222222	1	369.91499999999996
10	582	71.41006944444445	1	208.60500000000002
3	510	527.496875	1	144.64999999999998



(a) 3D Color-Coded Segments



(b) Cluster Median Plot

Figure 9: Final color-coded 3D customer segment clusters and corresponding median RFM summary visualizations. Click to view Color Coded Cluster(10) and Color Coded Cluster(13;rejected)

6 Business Interpretation and Strategy

6.1 Customer Archetypes and Segment Profiles

Based on the median RFM metrics derived from the optimal 10-component `VEV` Gaussian Mixture Model, we synthesize the clusters into four main operational personas:

- **Core Customers (Clusters 1, 2, 5, 7, 8):** Comprising 3,120 customers ($\approx 56\%$ of the total customer base), these buyers exhibit high recency ($33.9 \leq$ Median $R \leq 58.5$ days), frequent transactions ($4 \leq$ Median $F \leq 11.5$), and high spend (Median $M > £1,600$). Cluster 5 represents top VIP spenders ($£2,598.28$), while Cluster 8 represents power users with the highest transaction frequency (11.5 orders).

- **Recent New Buyers (Cluster 10):** Representing 582 customers with recent purchase activity ($R = 71.41$ days) but single transactions ($F = 1$) and low spend ($£208.61$). They reflect new acquisitions currently in the entry-level onboarding phase.

- **At-Risk Occasional Buyers (Cluster 6):** Consisting of 896 customers with moderate recency ($R = 214.87$ days) and low order frequency ($F = 2$). They are at immediate risk of slipping into churn if not re-engaged promptly.

- **High-Value Lost vs. Low-Value Churned Buyers (Clusters 3, 4, 9):** Cluster 4 ($n = 201$) displays high historical spending power ($£1,059.41$) despite severe inactivity ($R = 386.11$ days). Clusters 3 and 9 ($n = 1,034$) represent long-term inactive single-order buyers ($R >$ 420 days) with low overall spending.

6.2 Managerial Interventions

To maximize Customer Lifetime Value (CLV) and marketing ROI across these segments, we recommend targeted strategies:

1. **VIP Loyalty & Retention (Clusters 1, 2, 5, 7, 8):** Maintain engagement through dedicated account perks, priority customer service, and exclusive early access to product releases, avoiding unnecessary price-discounting that erodes margins.

2. **Onboarding & Basket-Expansion Triggers (Cluster 10):** Deploy automated post-purchase email sequences with personalized cross-category recommendations and threshold discounts (e.g., "£20 off on orders over £100") to encourage repeat orders.

3. **Targeted Win-Back for High-Value Lost Buyers (Cluster 4):** Prioritize high-impact re-engagement campaigns with strong personalized "We Miss You" incentives for Cluster 4, given their proven high historical spend ($£1,059.41$).

4. **Ad Spend Optimization for Low-Value Churned Buyers (Clusters 3 & 9):** Cease paid digital retargeting campaigns for Clusters 3 and 9 to conserve marketing budget, restricting future communications to low-cost bulk email sweeps.

A Appendices

A.1 Complete R Code Implementation Pipeline

Loading Libraries

Listing 1: Loading Necessary Libraries

```r
library(dplyr) #Data manipulation
library(readr) #Importing .csv or .feather files
library(arrow) #Writing lightweight .feather files
library(ggplot2) #2D plotting library
library(lubridate) #Date-time data parsing
library(plotly) #3D html interactive plotting library
library(mclust) #Clustering via GMM
library(purrr) #Functional manipulation in inertia calculation

#Most of the following code snippets are self sufficient, but
   loading these libraries will safeguard against missing libraries(
   if any)
```

Data Cleaning

Listing 2: Data Cleaning Pipeline

```r
library(dplyr)
library(readr)
df <- read_csv("Datasets/Online_retail_UCI.csv")
#Now we trim the missing Customer ID and cancelled orders
final_df<-df[!is.na(df$`Customer ID`) & !grepl("^C", as.character(df
   $Invoice)) ,]
write_csv(final_df, "Final_Online_Retail_UCI.csv")
```

Outlier Identification

Listing 3: Bivariate Quantity vs. Price Outlier Detection

```r
library(ggplot2)
library(readr)
df<-read_csv("Final_Online_Retail_UCI.csv")
# Wholesalers will be sorted here.
ggplot(df, aes(x=Price, y=Quantity, color=Country))+
  geom_point(size=3, alpha=0.5)

#After the violent outlier squish, we shall try narrowing down to
   the dense data region
ggplot(df, aes(x=Price, y=Quantity, color=Country))+
  geom_point(size=2, alpha=0.5)+
  coord_cartesian(xlim = c(0, 100), ylim = c(0, 200))+
  theme(legend.position = "none")
```

```
# The quantity cap is to ensure pure B2C transactions while the
   price cap takes care of the administrative entry issues, if any.
```

Retail Filtering

Listing 4: Retail-Only Consumer Isolation

```
df_retail <- df %>%
  filter(
    Quantity>0,
    Quantity<=200,
    Price>0,
    Price<=100
  )

write_csv(df_retail, "Retail_Only_Dataset.csv")
write_feather(df_retail, "Retail_Only_Dataset.feather")
```

Geographic Analysis

Listing 5: Countrywise Transaction Distribution

```
#Sorting number of transactions across the countries
top_countries <- df %>%
  group_by(Country) %>%
  summarise(Transactions = n()) %>%
  arrange(desc(Transactions))

library(arrow)
write_feather(top_countries, "Countrywise_Transactions.feather")
#Feather extensions are for fast loading (can't view, binary)
write_csv(top_countries,"Countrywise_Transactions_Dataset.csv")
#CSV files are for viewing

#For plotting, we shall take only the first 10 countries
top_10_countries<-top_countries %>% head(10)
ggplot(top_10_countries)+
  geom_bar(mapping=aes(x=reorder(Country, Transactions), y=
     Transactions), stat="identity", fill="greenyellow")+
  coord_flip()+labs(title="Countrywise Distributions of Transactions
     ")
#Note that this reordering is necessary as otherwise the countries
   will be plotted in random order
#See Country_transaction.csv attached
#The coord_flip is to ease the readability
```

Macro Revenue Analysis

Listing 6: Monthly Revenue Trend Calculation

```r
#Monthly Transactions
monthly_sales <- df %>%
  mutate(df, Revenue=df$Quantity*df$Price) %>%
  mutate(Month = floor_date(InvoiceDate, "month")) %>%
  group_by(Month) %>%
  summarise(TotalRevenue = sum(Revenue, na.rm = TRUE))
#2 new columns 'Revenue' and 'Month' are created.
write_csv(monthly_sales, "Monthly_Sales.csv")

#Data Visualization
library(ggplot2)
ggplot(monthly_sales, aes(x=Month, y=TotalRevenue))+
  geom_line(color="steelblue", linewidth=2)+
  geom_point(color="red", size=3)+
  labs(title = "Monthly Revenue Trend", x="Month", y="Revenue")
```

RFM Score Calculation

Listing 7: Recency, Frequency, and Monetary Aggregation

```r
library(lubridate)
df_retail<-read_feather("Retail_Only_Dataset.feather")

#Set a Reference Date
reference_date<-max(df_retail$InvoiceDate)+days(1) #To avoid Recency
    score=0

#Calculating the RFM score
RFM_score<- df_retail %>% group_by(df_retail$`Customer ID`) %>%
  summarise(
    #Recency score
    Recency= as.numeric(difftime(reference_date, max(InvoiceDate),
       units = "days")),

    #Frequency score
    Frequency=n_distinct(Invoice),
    # Simply adding up no. of entries won't work, we want no. of
       diff. transactions

    #Monetary score
    Monetary=sum(Quantity*Price)
  ) %>% ungroup() # To get rid of the grouping metadata

write_csv(RFM_score, "RFM_score.csv")
write_feather(RFM_score, "RFM_score.feather")
```

RFM Log Transformation

Listing 8: Natural Logarithmic Scaling of RFM Metrics

```r
#Logarithm helps us to smoothen the right skewed nature of the data
```

```
RFM_log<- RFM_score %>%
  mutate(
    log_R=log(Recency),
    log_F=log(Frequency),
    log_M=log(Monetary)
  )

#Plotting the log transformed RFM metrics
plot_ly(
  data=RFM_log,
  x=~log_R,
  y=~log_F,
  z=~log_M
)
```

The reasons for which we did not add 1 inside log unlike the report are as follows:

- Frequency can't be 0 as we have got rid of all the missing Customer IDs and canceled transactions.
- By our definition Recency score ≥ 1. Refer to RFM score code snippet.
- Due to Retail filtering boundary definition, Monetary score will also be strictly positive.

RFM Standardization

Listing 9: Z-Score Standardization of RFM Features

```
#To control the spread of the data along different axes
RFM_scaled <- RFM_log %>%
  mutate(
    z_R=as.numeric(scale(log_R)),
    z_F=as.numeric(scale(log_F)),
    z_M=as.numeric(scale(log_M))
  )
```

3D Scatter Plots

Listing 10: Initial 3D Scatter Plot Visualization

```
library(plotly)

#Initial 3D plot
plot_ly(
  data = RFM_score,
  x=~Recency,
  y=~Frequency,
  z=~Monetary
)

#Standardized RFM score plot
plot_ly(
  data=RFM_scaled,
```

```
  x=~z_R,
  y=~z_F,
  z=~z_M
)
```

GMM Clustering & BIC Selection

Listing 11: Model Fitting and BIC Selection Curve Evaluation

```
library(mclust)

# Gaussian Mixture Model Fitting
model_data <- RFM_scaled %>% select(z_R, z_F, z_M)
set.seed(0) # To initialize the EM and for reproducibility
gmm_fit <- Mclust(model_data, G=2:20)

#Viewing the BIC change
plot(gmm_fit, what = "BIC")#Grab the uniformly better model(VEV)
plot(gmm_fit, what = "BIC", legendArgs=list(plot=FALSE))
```

Optimal Model Fitting

Listing 12: Final Model Fitting with MAP Classification

```
set.seed(0)
gmm_fit_10 <- Mclust(model_data, G=10, modelNames = "VEV")

#Attaching the cluster labels
RFM_clustered_10 <- RFM_scaled %>%
  mutate(
    Cluster=as.factor(gmm_fit_10$classification),
    Uncertainty=gmm_fit_10$uncertainty
  )
#Assign each point to its most probable cluster
```

Post-Cluster Analysis

Listing 13: Cluster Behavioral Profiling Statistics

```
Cluster_EDA_10 <- RFM_clustered_10 %>%
  group_by(Cluster) %>%
  summarise(
    Customer_count=n(),
    Median_Recency= median(Recency),
    Median_Frequency= median(Frequency),
    Median_Monetary= median(Monetary)
  ) %>%
  arrange(desc(Median_Monetary))

write_csv(Cluster_EDA_10, "Cluster_EDA_10.csv")
```

Cluster Visualizations

Listing 14: Cluster EDA Plot Visualization

```r
library(plotly)

#Cluster EDA plot
plot_ly(
  data=Cluster_EDA,
  x=~Median_Recency,
  y=~Median_Frequency,
  z=~Median_Monetary
)
```

Color-Coded Segment Plot

Listing 15: 3D Color-Coded Segment Visualization

```r
#Color-coded plot
library(plotly)
plot_ly(
  data=RFM_clustered_10,
  x = ~z_R,
  y = ~z_F,
  z = ~z_M,
  color = ~Cluster,
  colors = "Set1",
  marker=list(size=3, opacity=0.6)
) %>%
  layout(
    showlegend = FALSE,
    scene = list(
      xaxis = list(title = "Scaled log(Recency)"),
      yaxis = list(title = "Scaled log(Frequency)"),
      zaxis = list(title = "Scaled log(Monetary)")
    )
  )
```

Silhouette Index Calculation

Listing 16: Calculating the Silhouette Index

```r
library(cluster)
#Poor performance of traditional hard clustering metrics
sil_idx_13 <- silhouette(gmm_fit_13$classification, dist(model_data)
   )
print(summary(sil_idx_13)$avg.width)
plot(sil_idx_13, main="Silhouette Plot for #Clusters=13", col=1:13)
#Use sil_idx_10 for the clusters 10.
# obtained avg widths are :C13=-0.0285, C10=0.00996
```

Elbow Method for Hard Clustering

Listing 17: The Elbow plot for Inertia

```r
library(ggplot2)
library(purrr)

# define a function to compute inertia (total within-cluster sum of
    squares)
compute_inertia <- function(k) {
  # nstart = 25 ensures stable centroid initialization
  kmeans(model_data, centers = k, nstart = 25)$tot.withinss
}

# Evaluate K from 1 to 15
k_values <- 1:15
set.seed(42) # Set seed for reproducibility
inertia_values <- map_dbl(k_values, compute_inertia)

# create a dataframe for ggplot
elbow_data <- data.frame(K = k_values, Inertia = inertia_values)

# Generate the Elbow Plot
ggplot(elbow_data, aes(x = K, y = Inertia)) +
  geom_line(color = "steelblue", size = 1) +
  geom_point(color = "darkred", size = 3) +
  scale_x_continuous(breaks = 1:15) +
  labs(
    title = "Elbow Method: Inertia (WCSS) vs. Number of Clusters",
    x = "Number of Clusters (K)",
    y = "Total Within-Cluster Sum of Squares (Inertia)"
  ) +
  theme_minimal()
```

A.2 Model Fitting Outputs

Run summary(gmm_fit_10) and 13 to get the following outputs.

Model Summary for 13 Clusters

Listing 18: Model Summary for 13 Clusters

```
----------------------------------------------------
Gaussian finite mixture model fitted by EM algorithm
----------------------------------------------------

Mclust VEV (ellipsoidal, equal shape) model with 13 components:

 log-likelihood    n  df         BIC         ICL
        -12840.9 5833 105 -26592.28 -27908.88

```

```
Clustering table:
  1   2   3   4   5   6   7   8   9  10  11  12  13
178 107 505  44 627 113 907 757 748 526 328 585 408
```

Model Summary for 10 Clusters

Listing 19: Model Summary for 10 Clusters

```
----------------------------------------------------
Gaussian finite mixture model fitted by EM algorithm
----------------------------------------------------

Mclust VEV (ellipsoidal, equal shape) model with 10 components:

 log-likelihood    n df         BIC       ICL
      -13566.33 5833 81 -27835.04 -29043.8

Clustering table:
  1   2   3   4   5   6   7   8   9  10
677 871 510 201 345 896 977 250 524 582
```

These outputs supplement the section Model Selection.

A.3 Von Neumann Trace Inequality

The following result, a corollary of the original Von Neumann Trace Inequality was used in the Covariance Matrix optimization: For Hermitian $n \times n$ positive semi-definite complex matrices A and B where now the eigenvalues are sorted decreasingly ($a_1 \geq a_2 \geq \cdots \geq a_n$ and $b_1 \geq b_2 \geq \cdots \geq b_n$, respectively),

$$\sum_{i=1}^{n} a_i b_{n-i+1} \leq \mathrm{Tr}(AB) \leq \sum_{i=1}^{n} a_i b_i.$$

With the right equality holding true when the matrices have the same set of eigenvectors in the same order of the eigenvalues, and the left one while the same eigenvectors correspond to opposite order of eigenvalues (one increasing, the other decreasing).
This result was referred to in VEV Covariance Matrix Optimization.

References

[1] Banfield, J. D., & Raftery, A. E. (1993). Model-based Gaussian and non-Gaussian clustering. *Biometrics*, 49(3), 803–821.

[2] Scrucca L, Fop M, Murphy TB, Raftery AE. mclust 5: Clustering, Classification and Density Estimation Using Gaussian Finite Mixture Models. R J. 2016 Aug;8(1):289-317. PMID: 27818791; PMCID: PMC5096736.